FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Object Oriented Programming

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Subject and Section	**Date:**
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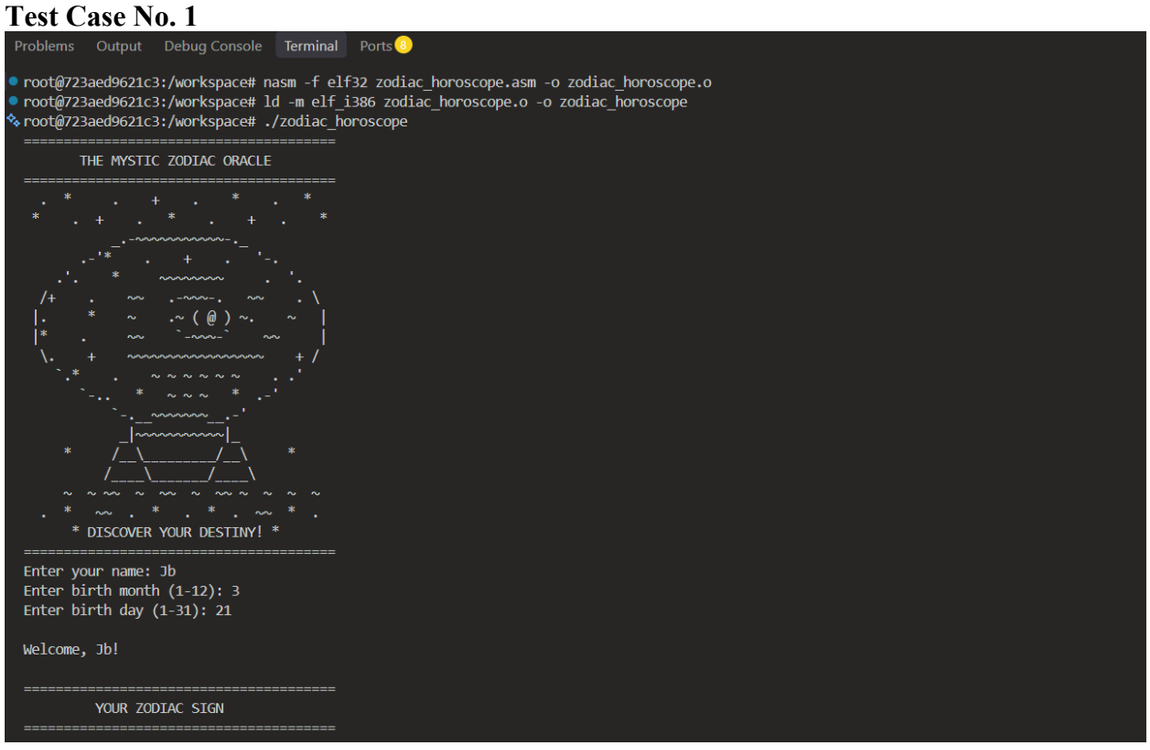
Machine Problem 1 — Zodiac Horoscope and Fortune Teller

I. Test Cases/Screenshots of Successful Execution/Invalid-input Handling

Test Cases

Test	Input	Expected result
1	March 21	Aries
2	April 19	Aries
3	April 20	Taurus
4	June 15	Gemini
5	July 23	Leo
6	August 15	Leo
7	December 25	Capricorn
8	February 30	Invalid date
9	Month 13	Invalid month
10	Menu choice 9	Invalid selection

Test Case No. 1





Test Case No. 2





```
                                                  '=xOOOox~'
                                    .':::`.        '*%@%XXO#%@#=.
                                  `=oX%%%%%%%%Xo+:  .:X@@O*+*oOX#@@+
                                 :o%@#X*x=;+-xxo#@@X#%@@@@XOoO%@%%#@#'
                                '*%%O*~+-xXOooOoO=+o+X@@O--;=xX%@@@@@#%%;
                               .o@%==o@@%%%##%@%#o*+O@x       `+#@@@@@%`
                               +@%xOOX%@@@###@@@@#X%@O;:-===+OXX@@@%@O -x+.
                               o@X=*O@@@O*oX#%%#%#o-`oXO*=*=;`'o@@@@-+%@%-
       '`.                    .O@o=+O@@X#@#OX%##XO;     -x*:     .*%@%@%@%;
    .=X%%#:                   x@%xXO@@Xo%%XOO#%@X*'  .o@%=       ~#@%%@%=
    x@#=%@~      ';+xxxoOO***+~-~x@@OxoX@@%XooOX%%#xX='  ':`.      . 'x@%x;
   `#@o.o@#=`-xX%@%OoOo*xX##X#%#X#@@#x*x#@@%#%@@@@o%X=:'         :X#ox%%-
  *@@X=:=O@%@%o=;` =x=`  ..;+:. .=X@@##XOooOO#@@O'*%%Xx-;`. .xO-;=X#%%-
  ~%@XO+-*@%*:=o*; =+' '   -*. .-; -o#@@@@%@@%O=` '-O%@%%##XooX###@@@*
   -#@@#X%%x:.*x:    `++.       `x*' ':;=**+~;.  =*:  :+O#%@@X#%%%%#o;
    ;O%@@%+==`:~' '  :x=        -=`    .'.`.    .:~;:~~*O#%@%- .'''
      '#@X=o=.   ~+`  ..   '.            .*+-.  ''  `=*~~=%@o
      `%@o-O*.   ;+`     .*o:     '=;     .``   -O=:'.`+oO@@+
      ;%@O`-+='       :;..oX*.    'oO='        ''~+;'=xxx%@@%#O*:
      `%@O~=oO*'     'oX+'.-+;     '--.        ~X*;-~=*+#@@#x*o%%*.
       *@@x::=x-   ';+%%**=:'. -*;' `':`.      ;X@OoOXX%@@@%#O=-@%`
     :+X@%+-    ':~+o%@Xx*OoX+`:xo=;ox`-*`.    =O%@%@@@@@@@@@@o-@%`
    =%%Ox-;~;-;:;-o#@@@%%#oOx+~***~~#@X**=-. ':' .;+o%@X=-:o@@oX@@o.
    X@+''~+*+xxOH%@@@@@@@@@%XOoooXOoo%@@#o*+**oOx*~`  `X@=:X@@%o%;
   -%@=~O%@@%@@@@@@@@@@%xx%%%@@%@@@%@@%##X%@x.'#@-O@@@@@~
  ;#@X+O@%~;:-%@@@@@%#Ox=:  ...:~=-~~~-`. `::'  ;#@x'o@o'~%@@@#;
 -o%@%*o@@+.-o%@##@@O'.                      ;O%@X-x@#:  `*o+'
;%@@@@X@%+.=@@@@x%@O`                        .O@@@#O@%-
:X@@@@%o;  *@@@@%X+'                         .o@@@@#x:
 .xOx+`    `++=~`                             .-*x~.
```

~ A R I E S ~

Zodiac Sign: ARIES
Dates: Mar 21 - Apr 19
Element: Fire
Symbol: Ram
Personality: Bold, energetic, and ready to lead.

==========
HOROSCOPE MENU
==========
[1] Love and Relationships
[2] Career and Studies
[3] Health and Wellness
[4] Lucky Number and Color
[5] View Zodiac Information
[6] Exit
Enter your choice: 2

CAREER HOROSCOPE
Aries, your creativity will help you
solve a difficult problem today.
Stay focused and confident!
Return to menu? (Y/N): N

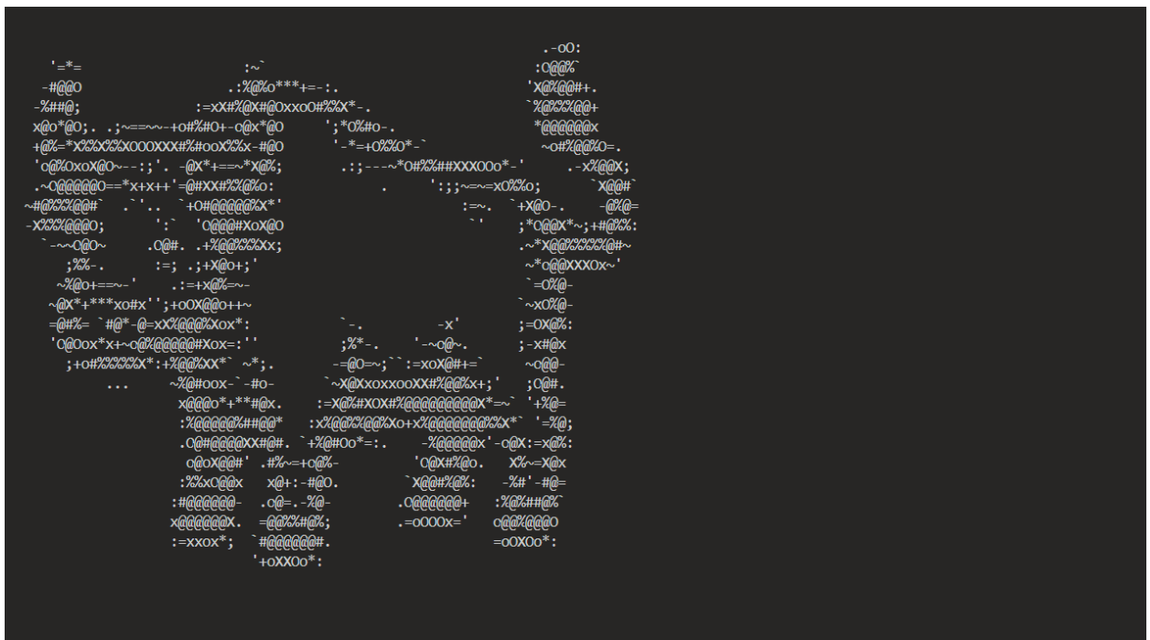
Thank you for consulting the Mystic Zodiac Oracle!
May the stars guide your path. Goodbye!

Test Case No. 3

```
root@723aed9621c3:/workspace# nasm -f elf32 zodiac_horoscope.asm -o zodiac_horoscope.o
root@723aed9621c3:/workspace# ld -m elf_i386 zodiac_horoscope.o -o zodiac_horoscope
root@723aed9621c3:/workspace# ./zodiac_horoscope
==========
       THE MYSTIC ZODIAC ORACLE
==========
   .  *      .    +      .     *      .   *
  *    .  +     .    *     .    +   .     *
           _.-~~~~~~~~~~~~~-._
        .-'*     .     +     .   '-.
     .'.    *      ~~~~~~~~~       .  '.
   /+     .     ~~    .-~~~~-.    ~~     . \
  |.      *     ~     .~ ( @ ) ~.     ~    |
  |*     .      ~~     `-~~~~-`     ~~     |
   \.    +     ~~~~~~~~~~~~~~~~~~~     + /
     `.*      .     ~ ~ ~ ~ ~ ~     . .'
       `-..     *    ~ ~ ~    *  .-'
            `-._ __    __ _.-'
               _|~~~~~~~~~~~|_
     *       /__\_________/__\       *
           /_____\_______/_____\
        ~  ~ ~~  ~  ~~  ~  ~~ ~  ~  ~  ~
   .  *   ~~  .  *    .  *  .  ~~  *  .
       * DISCOVER YOUR DESTINY! *
==========
Enter your name: Jb
Enter birth month (1-12): 4
Enter birth day (1-31): 20

Welcome, Jb!

==========
        YOUR ZODIAC SIGN
==========
```



```
                                                                .-oO:
    '=*=                    :~`                                 :O@@%`
   -#@@O                  .:%@%o***+=-:.                        'X@%@@#+.
  -%##@;              :=xX#%@X#@OxxoO#%%X*-.                    `%@%%%@@+
  x@o*@O;. .;~==~~-+o#%#O+-o@x*@O      ';*O%#o-.                  *@@@@@@x
  +@%=*X%%%%%XOOOXXX#%#ooX%%x-#@O      '-*=+O%%O*-`                ~o#%@@%O=.
  'o@%OxoX@O~--:;'. -@X*+==~*X@%;       .:;---~*O#%%##XXXOOo*-'      .-X%@@X;
  .~O@@@@@O==*X+X++'=@#XX#%%@%o:              .     ':;;~=~=xO%%o;       `X@@#`
~#@%%%@@#`  .`'..  `+O#@@@@@%X*'                    :=~.  `+X@O-.     -@%@=
-X%%%@@@O;       ':`  'O@@@#XoX@O                      `'    ;*O@@X*~;+#@%%:
  `-~~~O@O~     .O@#. .+%@@%%%Xx;                            .~*X@@%%%%@#~
     ;%%-.      :=; .;+X@o+;'                                ~*o@@XXXOx~'
    ~%@o+==~-'    .:=+x@%=~-                                  `=O%@-
   ^@X*+***xo#x'';+oOX@@o++~                                  `~xO%@-
   =@#%= `#@*-@=xX%@@@%Xox*:           `-.        -x'        ;=OX@%:
   'O@Oox*X+~o@%@@@@@#Xox=:''           ;%*-.    '-~o@~.      ;-x#@x
    ;+o#%%%%%X*:+%@@%XX*` ~*;.       -=@O=~;``:=xoX@#+=`     ~o@@-
         ...     ~%@#oox-`-#o-      `~X@XxoxxooXX#%@@%x+;'    ;O@#.
                 x@@@o*+**#@x.    :=X@%#XOX#%@@@@@@@@X*=~` '+%@=
                 :%@@@@@%##@@*   :x%@@%%@@%Xo+x%@@@@@@@@%X*`  '=%@;
                 .O@#@@@@XX#@#. `+%@#Oo*=:.    -%@@@@@x'-o@X:=x@%:
                  o@oX@@#' .#%~=+o@%-          'O@Xx#%@o.   X%~=X@x
                 :%%xO@@x   x@+:-#@O.         `X@@#%@%:   -%#'-#@=
                :#@@@@@@-  .o@=.-%@-         .O@@@@@@+   :%@%##@%`
                x@@@@@@X.  =@@%%#@%;         .=oOOOx='   o@@%@@@O
                :=xxox*;  `#@@@@@@#.                     =oOXOo*:
                          '+oXXOo*:
```

```
                          ~ T A U R U S ~
Zodiac Sign: TAURUS
Dates: Apr 20 - May 20
Element: Earth
Symbol: Bull
Personality: Patient, reliable, and devoted.

====================================
           HOROSCOPE MENU
====================================
[1] Love and Relationships
[2] Career and Studies
[3] Health and Wellness
[4] Lucky Number and Color
[5] View Zodiac Information
[6] Exit
Enter your choice: 3

HEALTH AND WELLNESS HOROSCOPE
Taurus, ground yourself with routine.
Regular meals, steady sleep and
time outdoors will restore balance.
Return to menu? (Y/N): N

Thank you for consulting the Mystic Zodiac Oracle!
May the stars guide your path. Goodbye!
```

Test Case No. 4

```
root@723aed9621c3:/workspace# nasm -f elf32 zodiac_horoscope.asm -o zodiac_horoscope.o
root@723aed9621c3:/workspace# ld -m elf_i386 zodiac_horoscope.o -o zodiac_horoscope
root@723aed9621c3:/workspace# ./zodiac_horoscope
====================================
       THE MYSTIC ZODIAC ORACLE
====================================
  .  *      .    +     .    *     .   *
 *     .  +     .   *      .     +   .    *
           _.-~~~~~~~~~~~~~~~~~~~~~~~~~~~~~-._
        .-'*     .     +     .     '-.
     .'.     *      ~~~~~~~~~~~~~~~      .  '.
   /+     .     ~~    .-~~~~~-.    ~~     . \
  |.      *     ~     .~ ( @ ) ~.     ~     |
  |*     .      ~~     `-~~~~-`      ~~     |
   \.     +     ~~~~~~~~~~~~~~~~~~~~~~~     + /
    `.*      .     ~ ~ ~ ~ ~ ~ ~     . .'
      `-..    *    ~ ~ ~    *   .-'
           `-._ ~~~~~~~~~~~~~ __.-'
              _|~~~~~~~~~~~~~|_
     *      /__\_____________/__\      *
           /____\_____________/____\
      ~   ~ ~~  ~   ~~  ~   ~~ ~  ~   ~  ~
   .  *    ~~   .    *     *   .   ~~   *   .
       * DISCOVER YOUR DESTINY! *
====================================
Enter your name: Jb
Enter birth month (1-12): 6
Enter birth day (1-31): 15

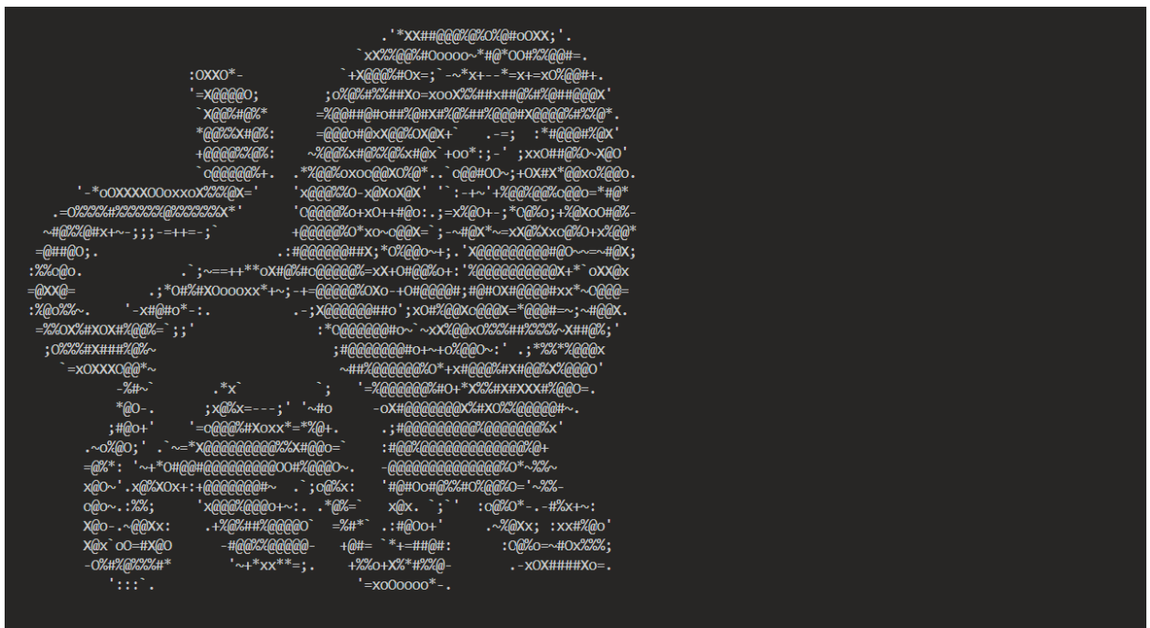
Welcome, Jb!

====================================
         YOUR ZODIAC SIGN
====================================
```



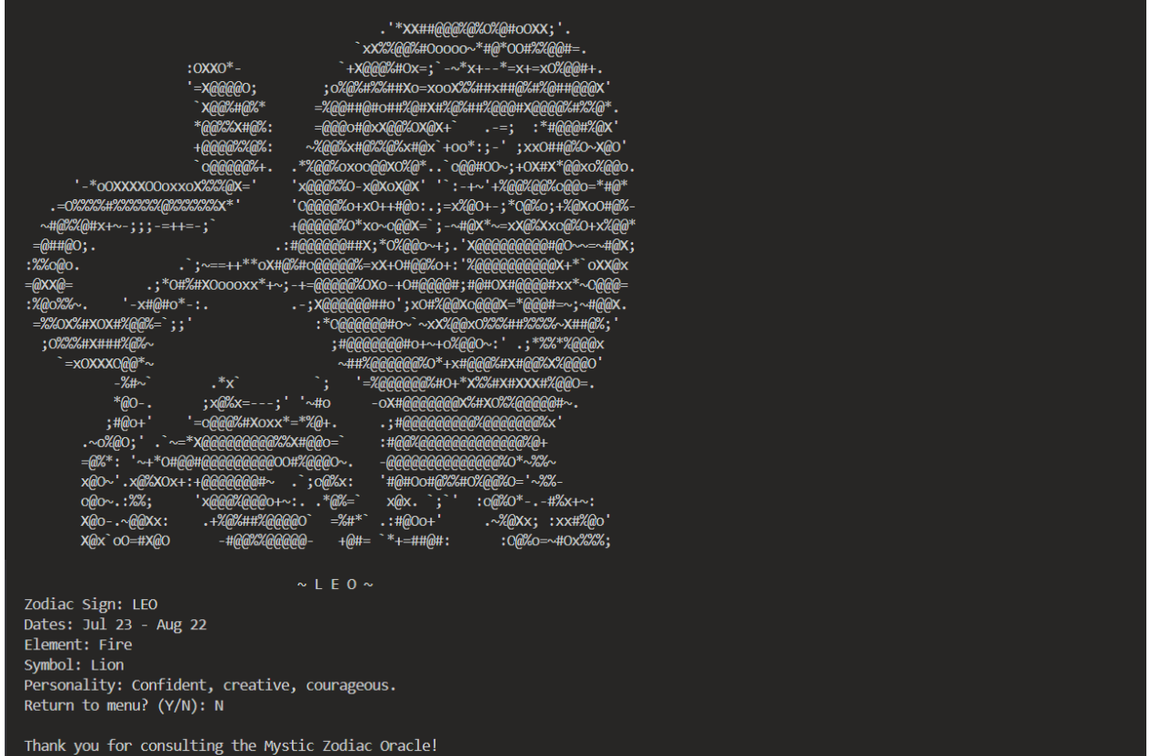
Test Case No. 5





```
                                    ~ L E O ~
Zodiac Sign: LEO
Dates: Jul 23 - Aug 22
Element: Fire
Symbol: Lion
Personality: Confident, creative, courageous.

==================================
          HOROSCOPE MENU
==================================
[1] Love and Relationships
[2] Career and Studies
[3] Health and Wellness
[4] Lucky Number and Color
[5] View Zodiac Information
[6] Exit
Enter your choice: 5
==================================
         YOUR ZODIAC SIGN
==================================
```



```
                                    ~ L E O ~
Zodiac Sign: LEO
Dates: Jul 23 - Aug 22
Element: Fire
Symbol: Lion
Personality: Confident, creative, courageous.
Return to menu? (Y/N): N

Thank you for consulting the Mystic Zodiac Oracle!
May the stars guide your path. Goodbye!
```

Test Case No. 6

```
root@723aed9621c3:/workspace# nasm -f elf32 zodiac_horoscope.asm -o zodiac_horoscope.o
root@723aed9621c3:/workspace# ld -m elf_i386 zodiac_horoscope.o -o zodiac_horoscope
root@723aed9621c3:/workspace# ./zodiac_horoscope
  ======================================
         THE MYSTIC ZODIAC ORACLE
  ======================================
    .  *      .     +     .     *     .    *
   *     .  +     .    *     .    +    .     *
               _.-~~~~~~~~~~~~~-._
          .-'*     .     +      .    '-.
       .'.     *      ~~~~~~~~~       .  '.
     /+     .     ~~    .-~~~~-.    ~~     . \
    |.      *     ~     .~ ( @ ) ~.     ~    |
    |*      .      ~~     `-~~~~-`     ~~     |
     \.     +     ~~~~~~~~~~~~~~~~~     + /
       `.*     .     ~ ~ ~ ~ ~ ~     . .'
          `-..    *    ~ ~ ~    *  .-'
              `-.__ ~~~~~~~ __.-'
                  _|~~~~~~~~~~~|_
        *       /___\_________/___\       *
              /_____\_______/_____\
       ~  ~ ~~  ~  ~~  ~  ~~ ~  ~  ~  ~
    .  *   ~~  .  *   .  *  .  ~~  *  .
         * DISCOVER YOUR DESTINY! *
  ======================================
  Enter your name: Jb
  Enter birth month (1-12): 8
  Enter birth day (1-31): 16

  Welcome, Jb!

  ======================================
           YOUR ZODIAC SIGN
  ======================================

                                          .'*XX##@@@%@%O%@#oOXX;'.
                                        `xX%%@@%#Ooooo~*#@*OO#%%@@#=.
                      :OXXO*-             `+X@@@%#Ox=;` ~~*X+--*=x+=xO%@@#+.
                      '=X@@@@O;          ;o%@%#%%##XO=xooX%%##x##@%#%@##@@@X'
                        `X@@%#@%*       =%@@##@#o##%@#X#%@%##%@@#X@@@@%#%%@*.
                       *@@%%%x#@%:      -@@@o#@@xX@@%OX@@X+`    .-=;  :*#@@@#%@X'
                      +@@@@%%@%:     ~%@@%x#@@%@%x#@x`+oo*:;-'  ;xxO##@%O~X@O'
                      `o@@@@@%+.   .*%@@%oxoo@@XO%@*..`o@@#OO~;+OX#X*@@xO%@@o.
        '-*oOXXXXOOoxxoX%%%@X='     'x@@@%%O-x@XoX@X' '`:-+~'+%@@%@@%o@@o=*#@*
     .=O%%%%#%%%%%@%%%%%%x*'        'o@@@@%o+xO++#@o:.;=x%@O+-;*O@%o;+%@xoO#@%-
   ~#@@%%@#x+~-;;;-=++=-;`          +@@@@@%O*xo~o@@x=`;-~#@X*~=xX@%Xxo@%O+x%@@*
  =@##@O;.                        .:#@@@@@@##X;*O%@@o~+;.'X@@@@@@@@@#@O~~=~#@x;
 :%%o@o.             .`;~==++**oX#@%#o@@@@@%-xX+O#@@%o+:'%@@@@@@@@@X+*`oXX@x
=@XX@=         .;*O#%#XOooxx*+~;-+=@@@@@%OXo-+O#@@@@#;#@#OX#@@@@#xx*~O@@@=
:%@o%%~.    '-x#@#o*-:.           .-;X@@@@@@##o';xO#%@@Xo@@@X=*@@@#=~;~#@@X.
 =%%OX%#XOX#%@@%=`;;'              :*O@@@@@@#o~`~xX%@@xO%%%##%%%~X##@%;'
  ;O%%%#X###%@%~                     ;#@@@@@@#o+~+o%@@O~:' .;*%%*%@@@x
    `=xOXXXO@@*~                      ~##%@@@@@%O*+x#@@@%#X#@@@X%@@@O'
          -%#~`       .*x`           `;    '=%@@@@@@%#O+*X%%#X#XXX#%@@O=.
          *@O-.      ;x@%x=---;' '~#o      -oX#@@@@@@@X%#XO%%@@@@@#~.
         ;#@o+'    '=o@@@%#XOxx*=*%@+.      .;#@@@@@@@@@%@@@@@@@%x'
       .~o%@o;' .`~=*X@@@@@@@@@%%X#@@o=`     :#@@%@@@@@@@@@@@@%@+
       =@%*: '~+*O#@@#@@@@@@@@@OO#%@@@O~.    -@@@@@@@@@@@@@@%O*~%%~
       x@O~'.x@%XOx+:+@@@@@@@#~  .`;o@%x:    '#@#Oo#@%%#O%@@%O='~%%-
       o@o~.:%%;      'x@@@%@@@o+~:. .*@%=`    x@x.  `;`  :o@%O*-.-#%x+~:
       X@o-.~@@Xx:    .+%@%##%@@@@O`   =%#*`  .:#@Oo+'     .~%@Xx; :xx#%@o'
       X@x`oO=#X@O      -#@@%%@@@@@-    +@#= `*+=##@#:       :O@%o=~#Ox%%%;
       -O%#%@@%%#*       '~+*xx**=;.     +%%o+x%*#%%@-       .-xOX####Xo=.
           ':::`.                        '=xoOoooo*-.



                                  ~ L E O ~
  Zodiac Sign: LEO
  Dates: Jul 23 - Aug 22
  Element: Fire
  Symbol: Lion
  Personality: Confident, creative, courageous.

  ======================================
             HOROSCOPE MENU
  ======================================
  [1] Love and Relationships
  [2] Career and Studies
  [3] Health and Wellness
  [4] Lucky Number and Color
  [5] View Zodiac Information
  [6] Exit
  Enter your choice: 1

  LOVE AND RELATIONSHIPS HOROSCOPE
  Leo, your passion is magnetic today.
  Speak from the heart and a warm
  connection will answer back.
  Return to menu? (Y/N): N

  Thank you for consulting the Mystic Zodiac Oracle!
  May the stars guide your path. Goodbye!
```

Test Case No. 7

```
root@723aed9621c3:/workspace# nasm -f elf32 zodiac_horoscope.asm -o zodiac_horoscope.o
root@723aed9621c3:/workspace# ld -m elf_i386 zodiac_horoscope.o -o zodiac_horoscope
root@723aed9621c3:/workspace# ./zodiac_horoscope
  ====================================
        THE MYSTIC ZODIAC ORACLE
  ====================================
    .  *      .     +     .     *     .   *
   *     .  +     .    *     .     +    .     *
                 _.-~~~~~~~~~~~~-._
            .-'*     .     +     .    '-.
         .'.      *      ~~~~~~~~~       .  '.
       /+      .     ~~    .-~~~-.    ~~     . \
       |.      *     ~     .~ ( @ ) ~.     ~    |
       |*     .      ~~     `-~~~-`     ~~      |
        \.     +     ~~~~~~~~~~~~~~~~~~     + /
          `.*      .     ~ ~ ~ ~ ~ ~     . .'
            `-..    *    ~ ~ ~    *   .-'
                 `-._ ~~~~~~~~ __.-'
                     _|~~~~~~~~~~|_
          *        /___\________/___\        *
                  /_____\______/_____\
          ~   ~ ~~   ~   ~~   ~   ~~ ~   ~   ~   ~
        .   *    ~~~   .   *    .   *   .   ~~~   *   .
            * DISCOVER YOUR DESTINY! *
  ====================================
  Enter your name: Jb
  Enter birth month (1-12): 12
  Enter birth day (1-31): 25

  Welcome, Jb!

  ====================================
           YOUR ZODIAC SIGN
  ====================================

  ====================================
           YOUR ZODIAC SIGN
  ====================================




                    ;~~:;%@%@%%##@#o+;``:-`
                  .X@%%%#oo%#%@@O+=;`
                   :o@%+:'=+=xO##%%@@-              :~`
                   ;x%~ .*x` ;#@@%#O~.            :O@@~                          ``
                 'x%o`  .==.'=x+O@o.              :%#O@x.                    :+X#*.
                 *@X*:: ':~~x%*';o@+              x@x+x%#-       :=-     -o%O~.
                 =@#X%xx%%@%#~   'O@x'            +@Oo++%#oOXOO#%@x   -X%o: .'..
                  -x#@x%%xx@O*;    .:x%%x:        'x%XX~##ox**oX%X`.*%o`.`:xox`
                    x@@#~O@+;.     .`'~o#X*-'       :%@%###XOoX%%*'.x@x~*O*xXOo`
                   :*+.'%#*=;        :;'~xX%%xo+;. ~%%X%@@%#O*~'  +@o.-ox~O#Xo:
                  -O#####@%#~.          ``     '-o@%#O%%XX@O:.      ;%O. -o~o#oo*:
                 '##+XXX#@@#OXO-             .+ooOoo%@%@X'     `xX%#: .*ooXoXOXO:
                 ;%o#%*oX%@@%X=.  `o~.  .~=*xO+OOOxO%@o.    'X%=;' '-x%%#Xo*~.
                 -@oX@x*+x%X='`=*x#@Ox+~~~=oxXOX*o*X*X@O`   +@o:~~+X@O=`   '``.
              :*oX@##@#+%#~~O#%##@@%%%#XX###X#XXooxOOOX@X' `%#:`oo#@x`=oO*xOo;
             ~%%o;*@@@#'O%~X@o='.`;+*xo%@@@@@@@%#oXOOXOX@= o%**+o@@o. '=o*ooo;
           *@o`*x'o#o:`;X%x#@@%o+;.  -@@@@@@@@%@@##oOo#@#o@%##OX@#`   ':=xoO#~
         .*@x; 'OO+*xO#%%@@xO@%OO%O: =@@@%%@@@OX@##XXXX@O~-;~*oX%#x-:=xoOXXX+
      :~+*O%xx~;~O#%%#XooXX%#@@XoX@%X#@@%#X%#%@%#oXOXX%@o*=`     .~o%#OOXOXO#=
    .x@Xx*=:.;+xOX%X+;'    :-o%%%#X%@@@%@@@%###XXOXOX#X%@#o*~`     `=*;o@%%###X:
   .x@Xx'-oOOO#%@@XXO+~*oox`:~o%%%#%@@Xoxo#@@#XX#O##%@@#o*=+;'  ..`=*o+x@@%@%X.
  :X%oo+`+oXX@@%+'.+X##o*O#=-~xO#@@X=--~*xxoX#%%####o=~+=:.'`'+Ox+XXXOo#@#%#X-
  =o-`~`.`:~-*x;   ;-:`..;*~-:==xX='.   .:'                  `=~:;;++~===~;~-'



  ====================================
             HOROSCOPE MENU
  ====================================
  [1] Love and Relationships
  [2] Career and Studies
  [3] Health and Wellness
  [4] Lucky Number and Color
  [5] View Zodiac Information
  [6] Exit
  Enter your choice: 2

  CAREER HOROSCOPE
  Capricorn, patient effort is about to pay off.
  Finish the task you keep postponing
  and a mentor will take notice.
  Return to menu? (Y/N): N

  Thank you for consulting the Mystic Zodiac Oracle!
  May the stars guide your path. Goodbye!
```

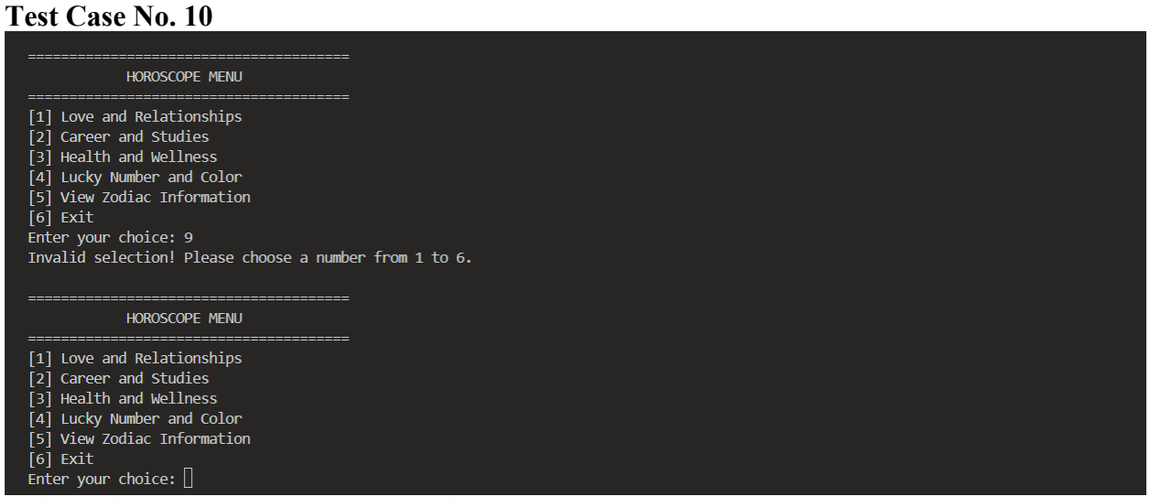
Test Case No. 8



Test Case No. 9



Test Case No. 10



II. Brief Explanation of Implemented Procedures

- The zodiac_horoscope.asm program uses separate procedures to organize the Zodiac Horoscope system. display_welcome presents the title and welcome ASCII art, while get_user_info collects the user's name, birth month, and birth day. validate_birthdate checks the month and day against valid calendar limits, including the number of days in each month.

The determine_zodiac procedure identifies the correct zodiac sign by comparing the birth month and day with the established zodiac date ranges. display_zodiac_art retrieves the appropriate artwork from art_table, while display_zodiac_info displays the sign name, date range, element, symbol, and personality description.

display_menu shows the horoscope choices, and show_horoscope displays the selected love, career, health, lucky-number, or zodiac-information category.

The supporting procedures read_line and parse_number handle user input, while print_string and print_number display text and numeric values. ask_return_menu validates whether the user wants to return to the menu, and exit_program terminates the application through the Linux int 0x80 system call. Together, these procedures provide organized input collection, date validation, zodiac identification, horoscope selection, menu repetition, and program termination

Machine Problem 2 — Dungeon Adventure and Survival Game

I. Test Cases/Screenshots of Successful Execution/Invalid-Input Handling

Test Cases

Test	Scenario	Expected result
1	Select Warrior	Health 100, Attack 20
2	Select Mage	Health 70, Attack 30
3	Select Archer	Health 85, Attack 25
4	Select class 9	Invalid selection
5	Warrior attacks Goblin twice	Goblin defeated
6	Potion at full health	Health capped; potion consumed
7	Potion count 0	Error; no health change
8	Open treasure	Gain 50 gold
9	Health reaches 0	Game Over
10	Dragon health reaches 0	Victory screen

Test Case No. 1

```
root@723aed9621c3:/workspace# nasm -f elf32 dungeon_game.asm -o dungeon_game.o
root@723aed9621c3:/workspace# ld -m elf_i386 dungeon_game.o -o dungeon_game
root@723aed9621c3:/workspace# ./dungeon_game
  ====================================
         DUNGEON OF THE LOST KING
  ====================================
               |>>>> |
             _ _|_ _ |
           |;|_|;|_|;|
           | \. . /
           \: . /
           ||: |
           ||:. |
           ||: .|
        ENTER IF YOU DARE!
  ====================================

  Enter your hero name: Princess
  Choose your character:
  [1] Warrior
  [2] Mage
  [3] Archer
  Enter choice: 1
  Welcome, Princess! Your class is Warrior

  ====================================
                MAIN MENU
  ====================================
  [1] Explore Dungeon
  [2] View Character Status
  [3] Use Health Potion
  [4] View Inventory
  [5] Exit Game
  Enter choice: 2


  ========== CHARACTER STATUS ==========
  Name: Princess
  Class: Warrior
  Health: 100 / 100
  Attack: 20
  Potions: 2
  Gold: 0
  Room: 1

  ====================================
                MAIN MENU
  ====================================
  [1] Explore Dungeon
  [2] View Character Status
  [3] Use Health Potion
  [4] View Inventory
  [5] Exit Game
  Enter choice: 5

  Thank you for playing.
root@723aed9621c3:/workspace#
```

Test Case No. 2

```
root@723aed9621c3:/workspace# nasm -f elf32 dungeon_game.asm -o dungeon_game.o
root@723aed9621c3:/workspace# ld -m elf_i386 dungeon_game.o -o dungeon_game
root@723aed9621c3:/workspace# ./dungeon_game
  ====================================
         DUNGEON OF THE LOST KING
  ====================================
              |>>>> |
            _ _|_ _ |
          |;|_|;|_|;|
          | \. . /
          \: . /
          ||: |
          ||:. |
          ||: .|
        ENTER IF YOU DARE!
  ====================================

  Enter your hero name: Princess
  Choose your character:
  [1] Warrior
  [2] Mage
  [3] Archer
  Enter choice: 2
  Welcome, Princess! Your class is Mage

  ====================================
                MAIN MENU
  ====================================
  [1] Explore Dungeon
  [2] View Character Status
  [3] Use Health Potion
  [4] View Inventory
  [5] Exit Game
  Enter choice: 2

  ========== CHARACTER STATUS ==========
  Name: Princess
  Class: Mage
  Health: 70 / 70
  Attack: 30
  Potions: 3
  Gold: 0
  Room: 1

  ====================================
                MAIN MENU
  ====================================
  [1] Explore Dungeon
  [2] View Character Status
  [3] Use Health Potion
  [4] View Inventory
  [5] Exit Game
  Enter choice: 5

  Thank you for playing.
root@723aed9621c3:/workspace#
```

Test Case No. 3

```
root@723aed9621c3:/workspace# nasm -f elf32 dungeon_game.asm -o dungeon_game.o
root@723aed9621c3:/workspace# ld -m elf_i386 dungeon_game.o -o dungeon_game
root@723aed9621c3:/workspace# ./dungeon_game
  ====================================
         DUNGEON OF THE LOST KING
  ====================================
              |>>>> |
            _ _|_ _ |
          |;|_|;|_|;|
          | \. . /
          \: . /
          ||: |
          ||:. |
          ||: .|
        ENTER IF YOU DARE!
  ====================================

  Enter your hero name: Princess
  Choose your character:
  [1] Warrior
  [2] Mage
  [3] Archer
  Enter choice: 3
  Welcome, Princess! Your class is Archer

  ====================================
                MAIN MENU
  ====================================
  [1] Explore Dungeon
  [2] View Character Status
  [3] Use Health Potion
  [4] View Inventory
  [5] Exit Game
  Enter choice: 2
```

```
========== CHARACTER STATUS ==========
Name: Princess
Class: Archer
Health: 85 / 85
Attack: 25
Potions: 2
Gold: 0
Room: 1

======================================
              MAIN MENU
======================================
[1] Explore Dungeon
[2] View Character Status
[3] Use Health Potion
[4] View Inventory
[5] Exit Game
Enter choice: 5

Thank you for playing.
root@723aed9621c3:/workspace#
```

Test Case No. 4

```
root@723aed9621c3:/workspace# nasm -f elf32 dungeon_game.asm -o dungeon_game.o
root@723aed9621c3:/workspace# ld -m elf_i386 dungeon_game.o -o dungeon_game
root@723aed9621c3:/workspace# ./dungeon_game
  ====================================
         DUNGEON OF THE LOST KING
  ====================================
              |>>>> |
              _ _|_ _ |
           |;|_|;|_|;|
           | \. . /
           \: . /
           ||: |
           ||:. |
           ||: .|
        ENTER IF YOU DARE!
  ====================================

  Enter your hero name: Princess
  Choose your character:
  [1] Warrior
  [2] Mage
  [3] Archer
  Enter choice: 9
  Invalid selection. Please choose 1, 2, or 3.
  Choose your character:
  [1] Warrior
  [2] Mage
  [3] Archer
  Enter choice:
```

Test Case No. 5

```
root@723aed9621c3:/workspace# nasm -f elf32 dungeon_game.asm -o dungeon_game.o
root@723aed9621c3:/workspace# ld -m elf_i386 dungeon_game.o -o dungeon_game
root@723aed9621c3:/workspace# ./dungeon_game
  ====================================
         DUNGEON OF THE LOST KING
  ====================================
              |>>>> |
              _ _|_ _ |
           |;|_|;|_|;|
           | \. . /
           \: . /
           ||: |
           ||:. |
           ||: .|
        ENTER IF YOU DARE!
  ====================================

  Enter your hero name: Princess
  Choose your character:
  [1] Warrior
  [2] Mage
  [3] Archer
  Enter choice: 1
  Welcome, Princess! Your class is Warrior

  ======================================
                MAIN MENU
  ======================================
  [1] Explore Dungeon
  [2] View Character Status
  [3] Use Health Potion
  [4] View Inventory
  [5] Exit Game
  Enter choice: 1
```

```
========== ROOM 1: GOBLIN CHAMBER ==========
      ' '
    /(.-""-.)\
    |\ \/ \/ /|
    | \ / =. .= \ / |
     \ \   ^   / /
      \ '-----' /
   A GOBLIN APPEARS!
Enemy Health: 30
Your Health: 100

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 1

You attacked the enemy!

The enemy attacked you!
Enemy Health: 10
Your Health: 90

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 1

You attacked the enemy!

You defeated the enemy!

========= ROOM 2: TREASURE CHAMBER =========
      _________
     /________/|
     |  ____  | |
     | |  ?  | | |
     | |____| | |
     |________|/
[1] Open Chest
[2] Leave Chest
Enter choice: 2
You left the chest behind.

========== ROOM 3: DRAGON'S LAIR ==========
       / \\
      //\\
     |\__/|
     /0 0 \
    |  ^  |
     / \__/ \
    / \_________\/
   THE DRAGON AWAKENS!
Enemy Health: 60
Your Health: 90

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 4

You ran away and returned to the main menu.

=======================================
              MAIN MENU
=======================================
[1] Explore Dungeon
[2] View Character Status
[3] Use Health Potion
[4] View Inventory
[5] Exit Game
Enter choice: 5

Thank you for playing.
root@723aed9621c3:/workspace#
```

Test Case No. 6

```
root@723aed9621c3:/workspace# nasm -f elf32 dungeon_game.asm -o dungeon_game.o
root@723aed9621c3:/workspace# ld -m elf_i386 dungeon_game.o -o dungeon_game
root@723aed9621c3:/workspace# ./dungeon_game
  ======================================
         DUNGEON OF THE LOST KING
  ======================================
              |>>>> |
            _ _|_ _ |
          |;|_|;|_|;|
          | \. . /
          \: . /
          ||: |
          ||:. |
          ||: .|
        ENTER IF YOU DARE!
  ======================================

  Enter your hero name: Princess
  Choose your character:
  [1] Warrior
  [2] Mage
  [3] Archer
  Enter choice: 1
  Welcome, Princess! Your class is Warrior

  ======================================
               MAIN MENU
  ======================================
  [1] Explore Dungeon
  [2] View Character Status
  [3] Use Health Potion
  [4] View Inventory
  [5] Exit Game
  Enter choice: 3
  Your health is already full. The potion was consumed.
```

Test Case No. 7

```
root@723aed9621c3:/workspace# nasm -f elf32 dungeon_game.asm -o dungeon_game.o
root@723aed9621c3:/workspace# ld -m elf_i386 dungeon_game.o -o dungeon_game
root@723aed9621c3:/workspace# ./dungeon_game
  ======================================
         DUNGEON OF THE LOST KING
  ======================================
              |>>>> |
            _ _|_ _ |
          |;|_|;|_|;|
          | \. . /
          \: . /
          ||: |
          ||:. |
          ||: .|
        ENTER IF YOU DARE!
  ======================================

  Enter your hero name: Princess
  Choose your character:
  [1] Warrior
  [2] Mage
  [3] Archer
  Enter choice: 1
  Welcome, Princess! Your class is Warrior

  ======================================
               MAIN MENU
  ======================================
  [1] Explore Dungeon
  [2] View Character Status
  [3] Use Health Potion
  [4] View Inventory
  [5] Exit Game
  Enter choice: 3
  Your health is already full. The potion was consumed.
```

```
 ====================================
              MAIN MENU
 ====================================
 [1] Explore Dungeon
 [2] View Character Status
 [3] Use Health Potion
 [4] View Inventory
 [5] Exit Game
 Enter choice: 3
 Your health is already full. The potion was consumed.

 ====================================
              MAIN MENU
 ====================================
 [1] Explore Dungeon
 [2] View Character Status
 [3] Use Health Potion
 [4] View Inventory
 [5] Exit Game
 Enter choice: 3
 Error: You have no health potions left. Health unchanged.

 ====================================
              MAIN MENU
 ====================================
 [1] Explore Dungeon
 [2] View Character Status
 [3] Use Health Potion
 [4] View Inventory
 [5] Exit Game
 Enter choice:
```

Test Case No. 8

```
root@723aed9621c3:/workspace# nasm -f elf32 dungeon_game.asm -o dungeon_game.o
root@723aed9621c3:/workspace# ld -m elf_i386 dungeon_game.o -o dungeon_game
root@723aed9621c3:/workspace# ./dungeon_game
 ====================================
       DUNGEON OF THE LOST KING
 ====================================
            |>>>> |
          _ _|_ _ |
        |;|_|;|_|;|
        | \. . /
        \: . /
        ||: |
        ||:. |
        ||: .|
      ENTER IF YOU DARE!
 ====================================

 Enter your hero name: Princess
 Choose your character:
 [1] Warrior
 [2] Mage
 [3] Archer
 Enter choice: 1
 Welcome, Princess! Your class is Warrior

 ====================================
              MAIN MENU
 ====================================
 [1] Explore Dungeon
 [2] View Character Status
 [3] Use Health Potion
 [4] View Inventory
 [5] Exit Game
 Enter choice: 1
```

```
========== ROOM 1: GOBLIN CHAMBER ==========
       , ,
      /(.-""-.)\
      |\ \/ \/ /|
      | \ / =. .= \ / |
       \ \   ^   / /
        \ '-----' /
   A GOBLIN APPEARS!
Enemy Health: 30
Your Health: 100

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 1

You attacked the enemy!

The enemy attacked you!
Enemy Health: 10
Your Health: 90

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 1

You attacked the enemy!

You defeated the enemy!

========= ROOM 2: TREASURE CHAMBER =========
      _________
     /_________/|
     |  _____  | |
     | |  ?  | | |
     | |_____| | |
     |_________|/
[1] Open Chest
[2] Leave Chest
Enter choice: 1
You opened the chest and gained 50 gold!

========== ROOM 3: DRAGON'S LAIR ==========
       / \\
      //\\
     |\___/|
     /0 0 \
    |  ^  |
     / \__/ \
    / \__________\/
   THE DRAGON AWAKENS!
Enemy Health: 60
Your Health: 90

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 1

You attacked the enemy!

The enemy attacked you!
Enemy Health: 40
Your Health: 75

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 1

You attacked the enemy!

The enemy attacked you!
Enemy Health: 20
Your Health: 60

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 1

You attacked the enemy!

You defeated the enemy!

VICTORY!
CONGRATULATIONS, Princess!
You defeated the dragon and escaped!
Final Gold: 50
Remaining Health: 60

YOU ARE A HERO!

Thank you for playing.
root@723aed9621c3:/workspace#
```

Test Case No. 9

```
root@723aed9621c3:/workspace# nasm -f elf32 dungeon_game.asm -o dungeon_game.o
root@723aed9621c3:/workspace# ld -m elf_i386 dungeon_game.o -o dungeon_game
root@723aed9621c3:/workspace# ./dungeon_game
 ====================================
       DUNGEON OF THE LOST KING
 ====================================
            |>>>> |
          _ _|_ _ |
         |;|_|;|_|;|
         | \. . /
         \: . /
         ||: |
         ||:. |
         ||: .|
       ENTER IF YOU DARE!
 ====================================

 Enter your hero name: Princess
 Choose your character:
 [1] Warrior
 [2] Mage
 [3] Archer
 Enter choice: 1
 Welcome, Princess! Your class is Warrior

 ====================================
              MAIN MENU
 ====================================
 [1] Explore Dungeon
 [2] View Character Status
 [3] Use Health Potion
 [4] View Inventory
 [5] Exit Game
 Enter choice: 1

========== ROOM 1: GOBLIN CHAMBER ==========
       , ,
     /(.-""-.)\
     |\ \/ \/ /|
     | \ / =. .= \ / |
      \ \   ^   / /
       \ '-----' /
   A GOBLIN APPEARS!
Enemy Health: 30
Your Health: 100

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 2

You defended and reduced the next enemy attack by 5.

The enemy attacked you!
Enemy Health: 30
Your Health: 95

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 2
```

```
You defended and reduced the next enemy attack by 5.

The enemy attacked you!
Enemy Health: 30
Your Health: 90

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 2

You defended and reduced the next enemy attack by 5.

The enemy attacked you!
Enemy Health: 30
Your Health: 85

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 2

You defended and reduced the next enemy attack by 5.

The enemy attacked you!
Enemy Health: 30
Your Health: 80

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 2

You defended and reduced the next enemy attack by 5.

The enemy attacked you!
Enemy Health: 30
Your Health: 5

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 2

You defended and reduced the next enemy attack by 5.

The enemy attacked you!

GAME OVER
Your hero has fallen.
The dungeon remains undefeated.
BETTER LUCK NEXT TIME!

Thank you for playing.
root@723aed9621c3:/workspace#
```

Test Case No. 10

```
Thank you for playing!
root@723aed9621c3:/workspace# nasm -f elf32 dungeon_game.asm -o dungeon_game.o
root@723aed9621c3:/workspace# ld -m elf_i386 dungeon_game.o -o dungeon_game
root@723aed9621c3:/workspace# ./dungeon_game
  ====================================
         DUNGEON OF THE LOST KING
  ====================================
              |>>>> |
            _ _|_ _ |
          |;|_|;|_|;|
          | \. . /
          \: . /
          ||: |
          ||:. |
          ||: .|
        ENTER IF YOU DARE!
  ====================================

  Enter your hero name: Princess
  Choose your character:
  [1] Warrior
  [2] Mage
  [3] Archer
  Enter choice: 1
  Welcome, Princess! Your class is Warrior

  ====================================
               MAIN MENU
  ====================================
  [1] Explore Dungeon
  [2] View Character Status
  [3] Use Health Potion
  [4] View Inventory
  [5] Exit Game
  Enter choice: 1

 ========== ROOM 1: GOBLIN CHAMBER ==========
       , ,
      /(.-""-.)\
      |\ \/ \/ /|
      | \ / =. .= \ / |
       \ \   ^   / /
        \ '-----' /
    A GOBLIN APPEARS!
 Enemy Health: 30
 Your Health: 100

 [1] Attack
 [2] Defend
 [3] Use Potion
 [4] Run
 Enter choice: 1

 You attacked the enemy!

 The enemy attacked you!
 Enemy Health: 10
 Your Health: 90

 [1] Attack
 [2] Defend
 [3] Use Potion
 [4] Run
 Enter choice: 1

 You attacked the enemy!

 You defeated the enemy!
```

```
========= ROOM 2: TREASURE CHAMBER =========

      _________
     /_________/|
     |  _____  | |
     | |  ?  | | |
     | |_____| | |
     |_________|/
[1] Open Chest
[2] Leave Chest
Enter choice: 1
You opened the chest and gained 50 gold!

========== ROOM 3: DRAGON'S LAIR ==========
       / \\
      //\\
     |\__/|
     /0 0 \
    |  ^  |
     / \__/ \
    / \__________\/
   THE DRAGON AWAKENS!
Enemy Health: 60
Your Health: 90

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 1

You attacked the enemy!

The enemy attacked you!
Enemy Health: 40
Your Health: 75

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 1

You attacked the enemy!

The enemy attacked you!
Enemy Health: 20
Your Health: 60

[1] Attack
[2] Defend
[3] Use Potion
[4] Run
Enter choice: 1

You attacked the enemy!

You defeated the enemy!

VICTORY!
CONGRATULATIONS, Princess!
You defeated the dragon and escaped!
Final Gold: 50
Remaining Health: 60

YOU ARE A HERO!

Thank you for playing.
root@723aed9621c3:/workspace#
```

II. Brief Explanation of Implemented Procedures

- The dungeon.asm program is divided into procedures to organize the Dungeon Adventure game and make the code easier to understand. display_welcome shows the game title and ASCII art, while get_player_info collects the hero's name and validates the selected character class. initialize_character assigns the appropriate health, attack power, and potion values for the Warrior, Mage, or Archer.

display_main_menu presents the available game options, while explore_dungeon controls the player's movement through the Goblin Chamber, Treasure Chamber, and Dragon's Lair. The battle_enemy procedure manages combat choices, including attacking, defending, using potions, and running away. enemy_retaliates calculates enemy damage and determines whether the player remains alive. The use_potion procedure restores health while preventing health from exceeding the maximum and preventing potion counts from becoming negative.

The display_status and display_inventory procedures show the player's current statistics, including health, attack power, potions, gold, and room number. display_game_result displays either the victory or Game Over screen. Supporting procedures such as read_line, read_int, print_string, print_int, and print_newline handle input and output operations. Finally, exit_program terminates the game using the Linux int 0x80 system call. These procedures work together to provide structured menu navigation, combat, room progression, validation, and game-result handling.

Open-Ended Questions

1. What is the purpose of procedures in your application?
   - Procedures are used to divide each program into smaller sections that perform specific responsibilities. In the Zodiac Horoscope program, separate procedures handle the welcome screen, user information, date validation, zodiac identification, zodiac information, menu display, horoscope selection, input reading, number conversion, and program termination. For example, get_user_info collects the user's name and birth date, while validate_birthdate checks whether the entered date is valid. After validation, determine_zodiac identifies the sign, and display_zodiac_info presents the appropriate information and artwork. The Dungeon Adventure program follows the same procedural structure. The get_player_info procedure registers the hero and validates the selected class, while initialize_character assigns the correct health, attack, and potion values. The explore_dungeon procedure controls room progression, battle_enemy handles combat decisions, and use_potion manages health restoration and potion limits. Other procedures, such as display_status, display_inventory, and display_game_result, are responsible for displaying specific information. Separating these responsibilities makes both programs easier to read, debug, test, and modify because a change in one feature does not require rewriting the entire program.

2. How do CALL and RET control program execution?
   - The CALL instruction transfers program execution to a procedure while saving the address of the next instruction on the stack. This allows the procedure to perform a particular task and then return to the exact location where it was called. The RET instruction retrieves the saved return address and sends execution back to the calling procedure. For example, the Zodiac program uses call determine_zodiac after the birth date has been validated. Once the zodiac sign has been stored, the RET instruction returns control to the main program so it can display the zodiac information. In the Dungeon program, CALL and RET are used throughout the main menu and combat system. When the player selects Explore Dungeon, the program calls explore_dungeon. When the player enters a room with an enemy, explore_dungeon calls battle_enemy. During battle, the program may call use_potion, enemy_retaliates, print_string, or read_number. After each procedure finishes, RET returns the program to the appropriate point. This procedure-based control flow allows the programs to reuse the same routines multiple times, reducing duplicate code and improving the organization of the assembly programs.
3. Which conditional jump instructions did you use, and why?
   - Both programs use the CMP instruction to compare values and conditional jumps to make decisions based on the comparison results. In the Zodiac program, JE is used to identify the birth month and direct the program to the correct month branch. For example, the program compares the birth month with 1, 2, 3, and the other month values. Once the correct month is found, additional comparisons determine the zodiac sign based on the birth day. The program uses JLE when a day is less than or equal to a zodiac boundary, such as January 19 or March 20. It uses JGE for ranges such as December 22 and later, which belongs to Capricorn. The Zodiac program also uses JL and JG during validation. These instructions reject months below 1, months above 12, days below 1, and days above the allowed number of days in a month. In the Dungeon program, JE is used to select character classes, main-menu options, combat actions,

and chest choices. The program uses JLE to determine whether an enemy's health is zero or below, indicating that the enemy has been defeated. It uses JG to check whether the player's health remains above zero. These jumps control the complete flow of both programs, including validation, menu navigation, combat, room progression, victory, defeat, and exit conditions.

4. How does your program handle invalid user input?
   - The Zodiac program performs several levels of input validation. First, it checks whether the user's name is empty. If the name contains no valid characters, the program displays an error message and requests the name again. It then validates the birth month by checking whether it is between 1 and 12. The birth day is checked to ensure that it is at least 1 and not greater than 31. After these general checks, the program uses the days_in_month table to determine the correct maximum number of days for the selected month. This allows the program to reject dates such as April 31 and February 30. February 29 is also rejected because the program does not collect a birth year to determine whether the year is a leap year. The Dungeon program also validates several types of input. During registration, the player must choose a class from 1 to 3. The main menu accepts only choices from 1 to 5, while the combat menu accepts only choices from 1 to 4. The chest menu accepts only choices 1 and 2. If an invalid value is entered, the program displays an error message and uses a jump to return to the corresponding input loop. The game also validates changing values during execution. It prevents the player's health from going below zero, prevents health from exceeding the maximum health, and prevents the potion count from becoming negative. When the potion count is zero, the program displays an error and returns without changing the player's health.

5. What would be harder to maintain without procedures?
   - The Zodiac program stores important values in memory variables such as birth_month, birth_day, and sign_index. The month and day variables preserve the date entered by the user, while sign_index stores the numerical index of the identified zodiac sign. This index is used with pointer tables to retrieve the correct zodiac name, date range, symbol, personality description, element, artwork, lucky number, and horoscope message. For example, art_table contains the addresses of the twelve zodiac artwork sections. The program uses the sign index to locate the correct address and then displays the artwork through print_string. The Zodiac program also uses a menu loop that repeatedly displays the horoscope options. After the user selects a category, the program displays the requested information and asks whether the user wants to return to the menu. If the user chooses to continue, control jumps back to menu_loop. If the user chooses to exit, the program calls exit_program. This loop allows the user to view multiple horoscope categories without restarting the application. The Dungeon program uses variables such as health, max_health, attack, potions, gold, current_room, and enemy_health to maintain the game state. These variables change as the player explores rooms, fights enemies, uses potions, and collects treasure. The current_room variable identifies whether the player is in Room 1, Room 2, or Room 3. The player advances after defeating the Goblin and after completing the Treasure Chamber. The combat loop repeatedly displays the combat menu until the player attacks, defends, uses a potion, runs away, defeats the enemy, or loses all health. Both programs use loops and data tables to make their behavior dynamic. The Zodiac program uses tables to select information based on the sign index, while the Dungeon program uses stored variables and repeated combat loops to track the changing condition of the player and enemies. This combination of procedures, loops, conditional branches, and memory variables allows both assembly programs to function as interactive applications rather than simple one-time displays.

Machine Problem 1 — Zodiac Horoscope and Fortune Teller

1. Explain how your determine_zodiac procedure identifies the correct sign from the birth month and day. Describe at least two conditional comparisons.
   - The determine_zodiac procedure reads the values stored in birth_month and birth_day. It first compares the birth month with numbers from 1 to 11 using instructions such as CMP and JE. If none of those comparisons match, the program uses the December section. After identifying the month, the procedure compares the birth day with the appropriate boundary for that month. For example, in the January section, the program uses cmp ebx, 19 followed by jle .capricorn. If the day is 19 or lower, the sign is Capricorn; otherwise, the program jumps to Aquarius. In the March section, it uses cmp ebx, 20 and jle .pisces. Days from March 1 to March 20 are assigned to Pisces, while later dates are assigned to Aries. These comparisons allow the program to determine the correct zodiac sign based on the date boundaries.

2. How does your program handle zodiac date ranges that cross two months, such as Capricorn? Explain the branching logic.
   - Capricorn is handled through two separate branches because its date range begins in December and continues into January. In the January branch, the program compares the day with 19. If the January day is less than or equal to 19, the program jumps to .capricorn; otherwise, it jumps to .aquarius. The December branch handles the other part of Capricorn's range. It compares the day with 22 using cmp ebx, 22 and jge .capricorn. Therefore, December 22 through December 31 is recognized as Capricorn, while December 1 through December 21 are directed to Sagittarius. This branching logic correctly handles a zodiac sign whose date range crosses from one calendar year into the next.

3. Why did you divide input collection, date validation, zodiac identification, and horoscope display into separate procedures?
   - These tasks are separated so that each procedure has one clear responsibility. The get_user_info procedure collects the user's name, birth month, and birth day. The validate_birthdate procedure checks whether the entered month and day represent a valid date. The determine_zodiac procedure identifies the zodiac sign only after the date has been confirmed as valid. Finally, display_zodiac_info and show_horoscope display the appropriate zodiac information and horoscope choices. This separation improves the organization and reliability of the program. If there is an error in date validation, it can be corrected inside validate_birthdate without changing the zodiac-identification logic. Similarly, new horoscope messages can be added to show_horoscope without changing how the user's name or birth date is collected. The procedures also make the program reusable because tasks such as printing strings, reading input, and displaying information can be called multiple times.

4. Describe how your program rejects invalid dates such as February 30 and an invalid month such as 13.
   - The validate_birthdate procedure first checks whether the month is within the valid range of 1 to 12. It uses cmp eax, 1 with jl .bad_month to reject values below 1, and cmp eax, 12 with jg .bad_month to reject values above 12. Therefore, a month such as 13 is rejected and the program displays the invalid-month message. After validating the month, the procedure checks that the day is between 1 and 31. It then uses the days_in_month table, which stores the maximum number of days for each month. For February, the table contains 28, so when the player enters February 30, the program compares 30 with 28 and jumps to .bad_date. The program displays the month name, the number of valid days, and the retry message. February 29 is also rejected because the program does not collect a birth year for leap-year validation.

5. How does the program select a horoscope category after the user enters a menu choice? Identify the conditional jump instructions used.
   - After displaying the horoscope menu, the program reads the user's input and converts it into a number using parse_number. The value is stored in EAX, and the main menu logic first checks whether the value is 6. If it is 6, the program exits. Otherwise, the value is passed to show_horoscope. Inside show_horoscope, the program uses CMP and JE instructions to select the category. cmp eax, 1 followed by je .love selects Love and Relationships. cmp eax, 2 followed by je .career selects Career and Studies, while cmp eax, 3 and je .health select Health and Wellness. The program uses cmp eax, 4 and je .lucky for the Lucky Number and Color option, and cmp eax, 5 and je .info to display the complete zodiac information.

6. How did you implement the zodiac ASCII art? Explain how the correct design is selected and displayed.
   - The zodiac artwork is stored as separate null-terminated data blocks, such as art_aries, art_taurus, and art_gemini. Each block contains several lines of ASCII characters followed by a zero byte, allowing it to be printed as a string. The program then stores the addresses of these art blocks in the art_table. The sign_index variable identifies the selected zodiac sign using an index from 0 to 11. The display_zodiac_art procedure loads this index, multiplies it by four, and uses it to retrieve the corresponding address from art_table. It then calls print_string to display the selected design. This table-based approach prevents the program from needing a separate long conditional branch for every zodiac artwork.

7. What happens when a user enters an invalid menu option? Explain how control returns to the menu.
   - If the user enters a number outside the valid menu range, the comparisons in show_horoscope do not match options 1 through 5. The program then executes the invalid-selection branch, prints msg_inv_menu, clears EAX by using xor eax, eax, and returns to the caller. Back in menu_loop, the program checks the returned value with cmp eax, 0 and je menu_loop. Because the returned value is zero, control jumps back to the beginning of the menu loop. The horoscope menu is displayed again, allowing the user to enter another choice. This prevents invalid input from terminating the program or causing it to access an incorrect horoscope table.

8. If you added a new horoscope category, which procedures and conditional branches would you modify? Explain why.
   - The menu_text data block would first need a new option for the additional category. For example, a seventh option could be added after the existing View Zodiac Information option. The main menu logic would also need to be updated so that the new option is not mistaken for an invalid selection or the exit option. The show_horoscope procedure would require a new comparison and conditional jump, such as cmp eax, 6 followed by je .new_category. A new data label would need to contain the category heading and horoscope messages. If the new category depended on the zodiac sign or element, a new table similar to love_table, career_table, or health_table would be necessary. The procedure would use sign_index or the element value to select the correct message. The return logic would also need to be reviewed. The new category branch should display its content, set EAX to 1 like the other valid categories, and return to ask_return_menu. This ensures that the new category behaves consistently with the existing options and does not incorrectly trigger the invalid-menu branch or exit the program.

Machine Problem 2 — Dungeon Adventure and Survival Game

1. Explain how initialize_character assigns different health, attack, and potion values for Warrior, Mage, and Archer.
   - The initialize_character procedure uses the value stored in class_choice to determine which character statistics should be assigned. It compares the selected value with 1 and 2. If the value is 1, the program jumps to set_warrior, where the Warrior receives 100 health, 20 attack power, and 2 potions. If the value is 2, the program jumps to set_mage, assigning the Mage 70 health, 30 attack power, and 3 potions. If neither comparison is true, the program treats the character as an Archer and assigns 85 health, 25 attack power, and 2 potions. The procedure also initializes the player's gold to zero, sets the starting room to Room 1, and resets the game-over and victory flags.

2. Describe the procedures and conditional jumps used when the player chooses Attack, Defend, Use Potion, or Run.
   - The battle_enemy procedure displays the combat menu and uses read_int to receive the player's choice. The program compares the input in EAX with the values 1, 2, 3, and 4. The JE instruction transfers control to the corresponding action. For Attack, the program jumps to player_attack, subtracts the player's attack value from enemy_health, and then allows the enemy to retaliate if it survives. For Defend, the program jumps to player_defend, sets defend_flag to 1, and calls enemy_retaliates, which reduces the next enemy attack by five. When the player selects Use Potion, the program jumps to battle_potion and calls use_potion. This procedure checks whether potions are available and restores health without exceeding the maximum health. When the player selects Run, the program jumps to player_run, displays a message, and returns to the main menu without advancing the dungeon room. If the player enters an invalid option, the program displays an error message and jumps back to battle_loop, allowing the player to choose again.

3. How does your battle_enemy procedure determine whether the player or enemy has been defeated? Explain the comparisons.
   - The battle_enemy procedure checks the enemy's health at the beginning of each combat loop. If enemy_health is less than or equal to zero, the program jumps to enemy_won, sets the enemy health to zero, and returns from the procedure. After the player attacks, the program subtracts the player's attack power from the enemy's health and compares the result with zero using CMP and JLE. If the enemy's health is zero or below, it jumps to enemy_defeated_now, sets the enemy health to zero, displays the defeated message, and returns a victory result. The player's defeat is checked inside enemy_retaliates. The enemy's damage is subtracted from the player's health, and the program compares the result with zero. If the health is zero or below, the program explicitly sets the health to zero and sets game_over to 1. This prevents negative health from being displayed. The main loop later checks the game_over flag and calls display_game_result, which displays the Game Over screen.

4. Explain how you prevent health from exceeding the maximum and prevent potion use when the inventory is empty.
   - The use_potion procedure first compares the value of potions with zero. If the value is zero or negative, the program jumps to no_potion_available and displays an error message without decreasing the potion count or changing the player's health. This prevents the program from using potions that are not available and prevents the potion count from becoming negative. If a potion is available, the program decreases the potion count by one and adds twenty to the player's health. It then compares the new health value with max_health. If the new value exceeds the maximum, the program replaces it with the maximum health value. Therefore, a player with 90/100 health may be restored to 100/100, but a player can never have more health than the assigned

maximum. If health is already full, the potion is still consumed, but the health remains unchanged.

5. How is the current dungeon room tracked, and what conditions allow the player to advance to the next room?
   - The current dungeon room is stored in the room variable. The explore_dungeon procedure compares this variable with 1, 2, and 3 to determine which room should be entered. When the player defeats the Goblin in Room 1, the program checks whether enemy_health is zero or below and then changes room to 2. The player can then enter the Treasure Chamber. In Room 2, the player chooses whether to open or leave the chest. Regardless of the choice, the program changes room to 3, allowing the player to proceed to the Dragon's Lair. In Room 3, the player must defeat the Dragon for the victory condition to be activated. If the player runs from an enemy, the room value does not advance, allowing the encounter to be attempted again later.
6. What is the purpose of separating display_status, display_inventory, and battle_enemy into different procedures?
   - Separating these procedures gives each part of the program a clear responsibility. The display_status procedure presents the player's name, class, health, attack power, potion count, gold, and current room. The display_inventory procedure focuses only on inventory-related information, particularly the number of potions and amount of gold. The battle_enemy procedure handles combat actions, enemy health, player choices, retaliation, and combat outcomes. This separation improves readability and makes the program easier to maintain. For example, if the inventory display needs to be changed, only display_inventory needs to be modified. Likewise, combat rules can be updated inside battle_enemy and enemy_retaliates without affecting the character-status display. These procedures can also be called repeatedly from different parts of the program, which avoids duplicating the same assembly instructions.

7. How does your program preserve important register values across procedures and Linux system calls? Give one example.
   - The program preserves register values in procedures that use registers that must remain unchanged during a call. For example, print_string uses the Linux write system call, which requires EAX to contain the system-call number, EBX to contain the output descriptor, ECX to contain the string address, and EDX to contain the string length. Before using these registers, print_string pushes EAX, EBX, ECX, EDX, and ESI onto the stack. After the system call finishes, it restores them using POP instructions. The print_number procedure also pushes important registers before converting and displaying an integer. This ensures that the value being printed and the registers used by the calling procedure are not accidentally destroyed. However, registers such as EAX are intentionally used for return values in procedures like read_int, enemy_retaliates, and battle_enemy. In those cases, changing EAX is part of the procedure's expected behavior because it communicates the result back to the caller.

8. If you introduced a fourth room with a stronger enemy, what program data, procedures, and conditions would need to change?
   - To add a fourth room, the program would first need new data definitions in the .data section. These would include a title for Room 4, ASCII art for the stronger enemy, the enemy's combat messages if different, and possibly a new victory or reward message. A new room number, such as 4, would need to be recognized by explore_dungeon. The room progression logic would also need to change so that defeating the Dragon in Room 3 sets the room variable to 4 instead of immediately setting the victory condition. The program would also need a new section in the dungeon exploration procedure, such as enter_room_four. This procedure would display the fourth room's title and ASCII art, assign the stronger enemy's health and attack damage, reset the defense value, and

call battle_enemy. If the player defeats the new enemy, the program could set game_won to 1 and display the final victory message. If the player's health reaches zero, the existing Game Over logic could be reused without major changes because enemy_retaliates already checks for zero health. If the stronger enemy required a special ability, the battle_enemy or enemy_retaliates procedures would also need additional conditional branches. For example, the program could use a counter to make the enemy perform a stronger attack every third turn. A new variable could be added in the .bss section to store the enemy's special-attack state or turn count. The existing procedures, conditional jumps, and result flags could still be reused, allowing the fourth room to be added without redesigning the entire program.